Reasoning Inside The Box

Gentzen Calculi for Herbrand Logics

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MUGS
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Herbrand Structures

Models

- Semantically speaking, logic is based on models
- FO classical logic: FO structures
- Finite model theory: finite FO structures
- Herbrand logic: Herbrand FO structures

Herbrand Structures

- Herbrand structures = FO structures with a fixed domain
- The domain = the set of closed terms

Example

Language	Domain
$\langle 0, s \rangle$	$\{0, s(0), s(s(0)), s(s(s(0))), \ldots\}$
$\langle 0, 1, f \rangle$	$\{0, 1, f(0), f(1), f(f(1)), \ldots\}$
$\langle c, d \rangle$	$\{c, d\}$

Herbrand Logic

- Herbrand logic = the logic that is induced by Herbrand structures
- Studied in [Genesereth&Kao'15]

Advantages:

- Natural sub-class of structures
- Simpler than classical FOL
- $\mathbb{N}$ is axiomatizable
- TC is axiomatizable
- MM of SSLP is axiomatizable

Disadvantages:

- Not compact
- Not R.E.
- Inherently incomplete
- $\vdash$ is language-dependent
- No proof theory so far

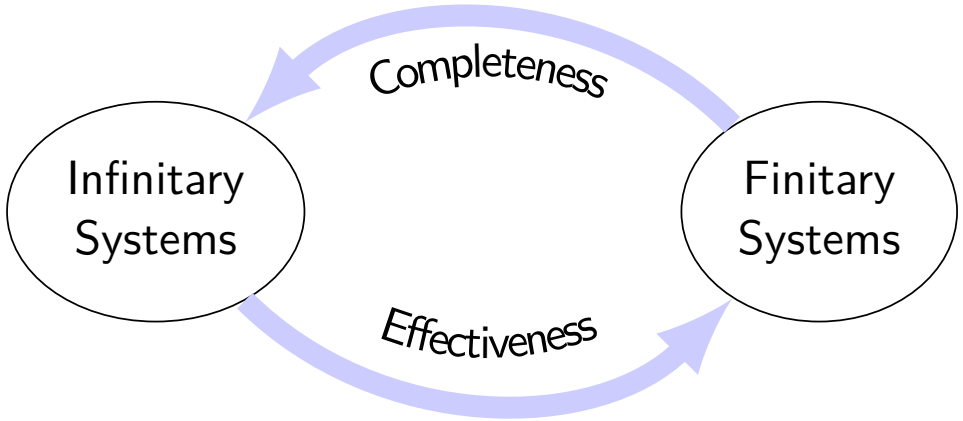
Main Results

Semantics:

- Decomposition of "Herbrandness"
- General treatment of equality

Proof Theory:

- Infinitary proof systems
  - Soundness
  - Completeness
- Finitary approximations
  - Soundness
  - (Completeness is impossible)



Herbrand Logics

Semantics

Semi-Herbrand Structures

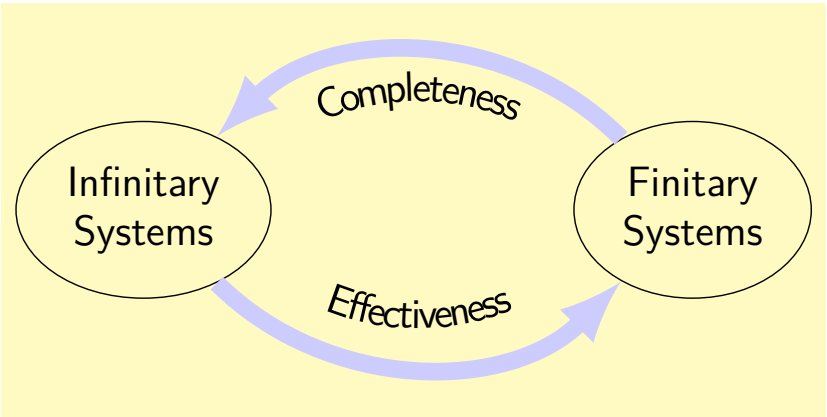
Herbrand Structures

Semi-Herbrand Structures with $=$

Herbrand Structures with $=$

Proof Systems

Infinitary Finitary



Herbrand and semi-Herbrand Structures

First Order Structures

- $M = \langle D, I \rangle$
- D is a set of elements (domain)
- I is an interpretation function for terms and predicates

Semi-Herbrand Structures

For every $d \in D$ there is $t \in \mathit{cterms}(\mathcal{L})$ s.t. $I(t) = d$

Herbrand Structures

For every $d \in D$ there is a unique $t \in \mathit{cterms}(\mathcal{L})$ s.t. $I(t) = d$

Herbrand Structures (equivalent definition)

- $D = \mathit{cterms}(\mathcal{L})$
- $I(t) = t$ for every $t \in \mathit{cterms}(\mathcal{L})$

Consequence Relations

Let $\mathcal{T}$ be a set of formulas and A a formula

- $\mathcal{T} \vdash_{cla} A$: every structure that satisfies $\mathcal{T}$ satisfies A
- $\mathcal{T} \vdash_{sHer} A$: every semi-Herbrand structure that satisfies $\mathcal{T}$ satisfies A
- $\mathcal{T} \vdash_{Her} A$: every Herbrand structure that satisfies $\mathcal{T}$ satisfies A

Are these relations different from one another?

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Are these relations different from one another?

Proposition

$\vdash_{cla} \subsetneq \vdash_{sHer} = \vdash_{Her}$

Thus, (semi-)Herbrand logic is super-classical.



Example

- $\{A\{\frac{t}{x}\} \mid t \in cterms(\mathcal{L})\} \nvdash_{cla} \forall x A$
- $\{A\{\frac{t}{x}\} \mid t \in cterms(\mathcal{L})\} \vdash_{Her} \forall x A$

Herbrand Logics

Semantics

Semi-Herbrand Structures



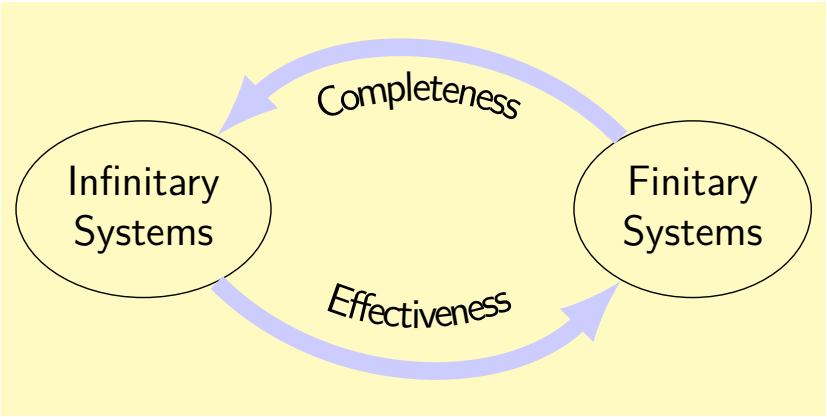
Herbrand Structures

Semi-Herbrand Structures with $=$

Herbrand Structures with $=$

Proof Systems

Infinitary Finitary



Equality

Suppose "$=$" is a predicate symbol of $\mathcal{L}$.

Normal Structures

A structure $M = \langle D, I \rangle$ is normal if $I(=)$ is $\{\langle a, a \rangle \mid a \in D\}$ ("$===$")

Consequence Relations

- $\mathcal{T} \vdash_{cla} A$: every structure that satisfies $\mathcal{T}$ satisfies A
- $\mathcal{T} \vdash_{sHer} A$: every semi-Herbrand structure that satisfies $\mathcal{T}$ satisfies A
- $\mathcal{T} \vdash_{Her} A$: every Herbrand structure that satisfies $\mathcal{T}$ satisfies A

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Consequence Relations

- $\mathcal{T} \vdash^{=}_{cla} A$: every normal structure that satisfies $\mathcal{T}$ satisfies A
- $\mathcal{T} \vdash^{=}_{sHer} A$: every normal semi-Herbrand structure that satisfies $\mathcal{T}$ satisfies A
- $\mathcal{T} \vdash^{=}_{Her} A$: every normal Herbrand structure that satisfies $\mathcal{T}$ satisfies A

Equality

Suppose "$=$" is a predicate symbol of $\mathcal{L}$.

Normal Structures

A structure $M = \langle D, I \rangle$ is normal if $I(=)$ is $\{\langle a, a\rangle \mid a \in D\}$ ("===")

Consequence Relations

- $\mathcal{T} \vdash^{=}_{cla} A$: every normal structure that satisfies $\mathcal{T}$ satisfies A
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- $\mathcal{T} \vdash^{=}_{Her} A$: every normal Herbrand structure that satisfies $\mathcal{T}$ satisfies A

Without Equality

$\vdash_{cla} \subsetneq \vdash_{sHer} = \vdash_{Her}$.

With Equality

$\vdash^{=}_{cla} \subsetneq \vdash^{=}_{sHer} \subsetneq \vdash^{=}_{Her}$

Example

$\vdash^{=}_{Her} c \neq d$, while $\nvdash^{=}_{sHer} c \neq d$.

Herbrand Logics

Semantics

Semi-Herbrand Structures

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Herbrand Structures

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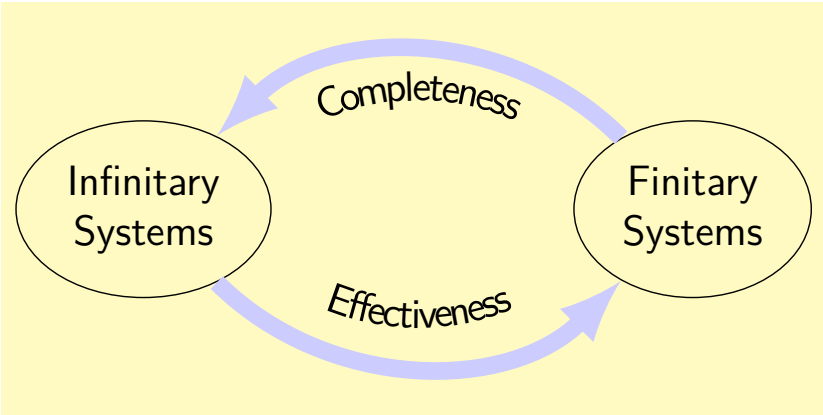
Semi-Herbrand Structures with $=$

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Herbrand Structures with $=$

Proof Systems

Infinitary Finitary



Axiomatization of Equality

Equality Formulas

- *Equiv* consists of the following formulas:
  - $x = x$
  - $x = y \supset y = x$
  - $(x = y \wedge y = z) \supset x = z$
- $Con(\mathcal{L})$ consists of the following formulas:
  - $(x_1 = y_1 \wedge \ldots \wedge x_n = y_n) \supset (f(x_1, \ldots, x_n) = f(y_1, \ldots, y_n))$
  - $(x_1 = y_1 \wedge \ldots \wedge x_n = y_n) \supset (P(x_1, \ldots, x_n) \supset P(y_1, \ldots, y_n))$

Fact

$\vdash^{=}_{cla} \quad \approx \quad \vdash_{cla}$

Question

Does the same hold for semi-Herbrand logic? Herbrand logic?

Yes. No.

Axiomatization of Equality

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Fact

$\mathcal{T} \vdash^{=}_{cla} A$ iff $\mathcal{T}, Equiv, Con(\mathcal{L}) \vdash_{cla} A$

Question

Does the same hold for semi-Herbrand logic? Herbrand logic?

Yes. No.

Equality in Herbrand Logics

semi-Herbrand Logic

$\mathcal{T} \vdash^{=}_{sHer} A$ iff $\mathcal{T}, Equiv, Con(\mathcal{L}) \vdash_{sHer} A$

Herbrand Logic

$\mathcal{T} \vdash^{=}_{cla} A$ ~~iff~~ $\mathcal{T}, Equiv, Con(\mathcal{L}) \vdash_{cla} A$

For Herbrand structures, inequalities are also needed (surprise surprise?)

Equality in Herbrand Logics

semi-Herbrand Logic

$\mathcal{T} \vdash^{=}_{sHer} A$ iff $\mathcal{T}, Equiv, Con(\mathcal{L}) \vdash_{sHer} A$

Herbrand Logic

$\mathcal{T} \vdash^{=}_{cla} A$ ~~iff~~ $\mathcal{T}, Equiv, Con(\mathcal{L}) \vdash_{cla} A$

Inequality Formulas

$inEq(\mathcal{L})$ consists of the following formulas:

- $f(x_1, \ldots, y_n) \neq g(y_1, \ldots, y_n)$ for every distinct f and g
- $x_i \neq y_i \supset f(\ldots, x_i, \ldots) \neq f(\ldots, y_i, \ldots)$

Herbrand Logic

$\mathcal{T} \vdash^{=}_{Her} A$ iff $\mathcal{T}, Equiv, Con(\mathcal{L}), inEq(\mathcal{L}) \vdash_{Her} A$

Equality in Herbrand Logics

semi-Herbrand Logic

$\mathcal{T} \vdash^{=}_{sHer} A$ iff $\mathcal{T}, Equiv, Con(\mathcal{L}) \vdash_{sHer} A$

Herbrand Logic

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Herbrand Logic

$\mathcal{T} \vdash^{=}_{Her} A$ iff $\mathcal{T}, x = x, inEq(\mathcal{L}) \vdash_{Her} A$

Example: Natural Numbers

Inequality Formulas

$inEq(\mathcal{L})$ consists of the following formulas:

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Herbrand Logic

$\mathcal{T} \vdash^{=}_{Her} A$ iff $\mathcal{T}, x = x, inEq(\mathcal{L}) \vdash_{Her} A$

Example (Relational PA without Induction)

$$\forall x.equal(x,x)$$
$$\forall x.(\neg equal(0,s(x)) \wedge \neg equal(s(x),0))$$
$$\forall x.\forall y.(\neg equal(x,y) \Rightarrow \neg equal(s(x),s(y)))$$

Example: Natural Numbers

Inequality Formulas

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Herbrand Logic

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The induction scheme is valid in Herbrand structures!

Herbrand Logics

Semantics

Semi-Herbrand Structures

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Herbrand Structures

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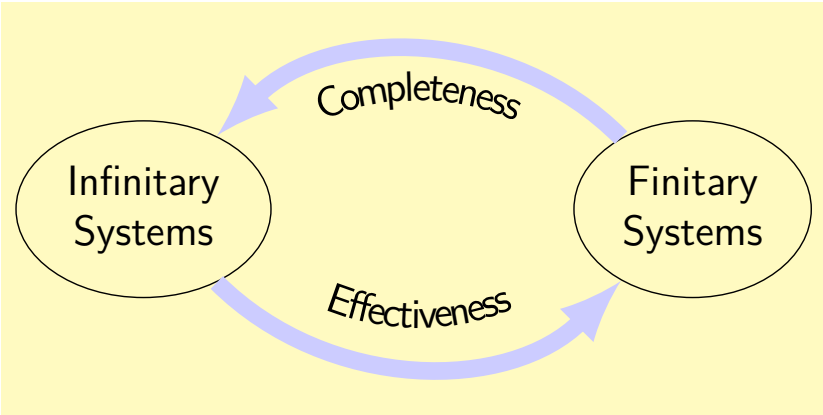
Semi-Herbrand Structures with $=$

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Herbrand Structures with $=$

Proof Systems

Infinitary Finitary



Herbrand Logics

Semantics

Thinking inside the box

No nameless elements



Proof Theory

Reasoning inside the box

No free variables



What Are Sequents?

- *Sequents* have the form $\Gamma \Rightarrow \Delta$, where Γ and Δ are finite sets of formulas.
- Intuition:

$$A_1, \ldots, A_n \Rightarrow B_1, \ldots, B_m \quad \rightsquigarrow \quad A_1 \wedge \ldots \wedge A_n \rightarrow B_1 \vee \ldots \vee B_m$$

- Special instance 1: Δ has one element: $\Gamma \Rightarrow A$
- Special instance 2: Γ is empty: $\Rightarrow A$

Example

- $A, B \Rightarrow A \wedge B$
- $A \Rightarrow A \vee B$
- $\Rightarrow A \vee \neg A$
- $A, \neg A \Rightarrow$
- $\Rightarrow A, \neg A$
- $A \Rightarrow A, B, C$

Sequent Calculus for Classical Logic **LK**

Γ and Δ are finite sets of formulas

Proof trees are finite

$$(id)\ \frac{}{\Gamma, A \Rightarrow A, \Delta} \qquad (cut)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \Delta}$$

$$(W \Rightarrow)\ \frac{\Gamma \Rightarrow \Delta}{\Gamma, A \Rightarrow \Delta} \qquad (\Rightarrow W)\ \frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow A, \Delta}$$

$$(\neg \Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, \neg A \Rightarrow \Delta} \qquad (\Rightarrow \neg)\ \frac{\Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \neg A, \Delta}$$

$$(\wedge \Rightarrow)\ \frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad (\Rightarrow \wedge)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}$$

$$(\vee \Rightarrow)\ \frac{\Gamma, A \Rightarrow \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \vee B \Rightarrow \Delta} \qquad (\Rightarrow \vee)\ \frac{\Gamma \Rightarrow A, B, \Delta}{\Gamma \Rightarrow A \vee B, \Delta}$$

$$(\supset \Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \supset B \Rightarrow \Delta} \qquad (\Rightarrow \supset)\ \frac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \supset B, \Delta}$$

Sequent Calculus for Classical Logic **LK**

Γ and Δ are finite sets of formulas

Proof trees are finite

$$(\forall \Rightarrow) \quad \frac{\Gamma, A\left\{\frac{t}{x}\right\} \Rightarrow \Delta}{\Gamma, \forall x A \Rightarrow \Delta}$$

$$(\Rightarrow \forall) \quad \frac{\Gamma \Rightarrow A\left\{\frac{y}{x}\right\}, \Delta}{\Gamma \Rightarrow \forall x A, \Delta} \text{ y is fresh}$$

$$(\exists \Rightarrow) \quad \frac{\Gamma, A\left\{\frac{y}{x}\right\} \Rightarrow \Delta}{\Gamma, \exists x A \Rightarrow \Delta} \text{ y is fresh}$$

$$(\Rightarrow \exists) \quad \frac{\Gamma \Rightarrow A\left\{\frac{t}{x}\right\}, \Delta}{\Gamma \Rightarrow \exists x A, \Delta}$$

Sequent Calculus for Herbrand Logic $\mathbf{G_{Her}}$

Γ and Δ are ~~finite~~ sets of closed formulas

Proof trees are of finite height

$$(\forall \Rightarrow) \quad \frac{\Gamma, A\left\{\frac{t}{x}\right\} \Rightarrow \Delta}{\Gamma, \forall x A \Rightarrow \Delta}\ t \text{ is closed}$$

~~$(\Rightarrow \forall)$~~ $\quad \dfrac{\Gamma \Rightarrow A\left\{\frac{y}{x}\right\}, \Delta}{\Gamma \Rightarrow \forall x A, \Delta}\ y \text{ is fresh}$

$$(\Rightarrow \forall)_H \quad \frac{\left\{\Gamma \Rightarrow A\left\{\frac{t}{x}\right\}, \Delta \mid t \in cterms(\mathcal{L})\right\}}{\Gamma \Rightarrow \forall x A, \Delta}$$

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Intuition Behind $\mathbf{G_{Her}}$

$$\frac{\left\{\Gamma \Rightarrow A\left\{\frac{t}{x}\right\}, \Delta \mid t \in cterms(\mathcal{L})\right\}}{\Gamma \Rightarrow \forall x A, \Delta}$$

ω-rule

- We have seen that:
  - $\left\{A\left\{\frac{t}{x}\right\} \mid t \in cterms(\mathcal{L})\right\} \nvdash_{cla} \forall x A$
  - $\left\{A\left\{\frac{t}{x}\right\} \mid t \in cterms(\mathcal{L})\right\} \vdash_{Her} \forall x A$
- This is known as the ω-rule
- It suffices to add this rule to characterize Herbrand semantics

Intuition Behind $\mathbf{G_{Her}}$

$$\frac{\left\{ \Rightarrow A\left\{\frac{t}{x}\right\} \mid t \in cterms(\mathcal{L})\right\}}{\Rightarrow \forall x A}$$

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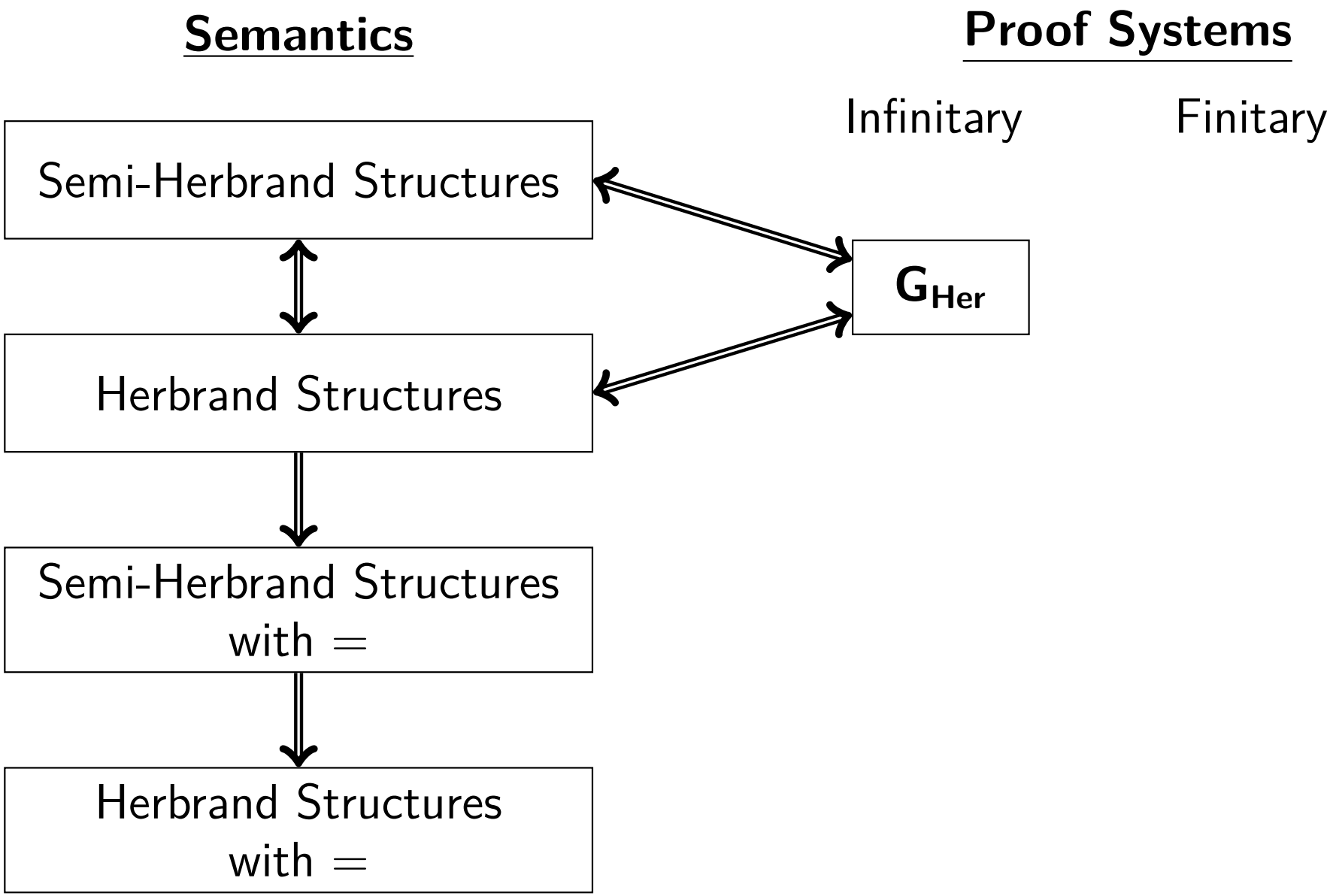
Theorem (soundness and completeness)

$\mathcal{T} \vdash_{Her} A$ iff the sequent $\mathcal{T} \Rightarrow A$ is derivable in $\mathbf{G_{Her}}$.

Bonus

The system is also complete for $\vdash_{sHer}$.

Herbrand Logics



Rules for Equality: Systems $\mathbf{G_{sHer_=}}$ and $\mathbf{G_{Her_=}}$

- $\mathbf{G_{sHer_=}} = \mathbf{G_{Her}} + (\Rightarrow=) + (\mathit{paramodulation})$
- $\mathbf{G_{Her_=}} = \mathbf{G_{Her}} + (\Rightarrow=) + (=\Rightarrow)$

$$(\mathit{paramodulation}) \quad \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, s = t \Rightarrow, A', \Delta} s, t \in \mathit{cterms}(\mathcal{L})$$

(A' is obtained from A by replacing s by t)

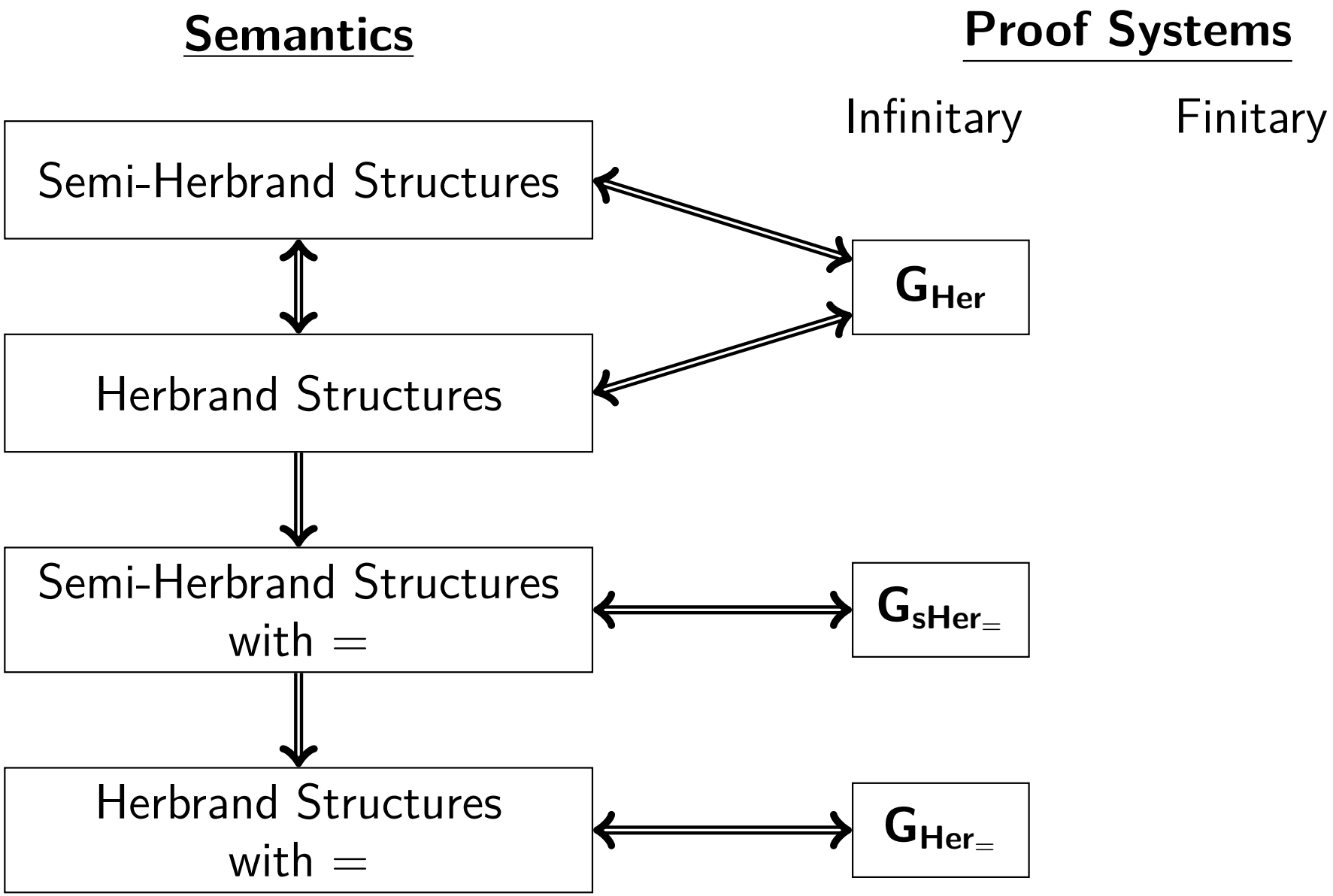
$$(\Rightarrow=) \quad \frac{}{\Gamma \Rightarrow t = t, \Delta} t \in \mathit{cterms}(\mathcal{L})$$

$$(=\Rightarrow) \quad \frac{}{\Gamma, s = t \Rightarrow \Delta} s \neq t \in \mathit{cterms}(\mathcal{L})$$

Theorem (soundness and completeness)

- $\mathcal{T} \vdash^{=}_{\mathit{sHer}} A$ iff the sequent $\mathcal{T} \Rightarrow A$ is derivable in $\mathbf{G_{sHer_=}}$.
- $\mathcal{T} \vdash^{=}_{\mathit{Her}} A$ iff the sequent $\mathcal{T} \Rightarrow A$ is derivable in $\mathbf{G_{Her_=}}$.

Herbrand Logics



Herbrand Logics

Semantics

Thinking inside the box

No nameless elements



Proof Theory

Reasoning inside the box

No free variables



Herbrand Logics

Semantics

Thinking inside the box

No nameless elements



Proof Theory

Peeking outside the box

free variables



Sequent Calculus for Herbrand Logic $\mathbf{G_{Her}}$

Γ and Δ are ~~finite~~ sets of closed formulas

Proof trees are of finite height

$$(\forall \Rightarrow) \quad \frac{\Gamma, A\left\{\frac{t}{x}\right\} \Rightarrow \Delta}{\Gamma, \forall x A \Rightarrow \Delta} \; t \text{ is closed}$$

~~$(\Rightarrow \forall)$~~ $\dfrac{\Gamma \Rightarrow A\left\{\frac{y}{x}\right\}, \Delta}{\Gamma \Rightarrow \forall x A, \Delta}$ y is fresh

$$(\Rightarrow \forall)_H \quad \frac{\left\{\Gamma \Rightarrow A\left\{\frac{t}{x}\right\}, \Delta \mid t \in cterms(\mathcal{L})\right\}}{\Gamma \Rightarrow \forall x A, \Delta}$$

~~$(\exists \Rightarrow)$~~ $\dfrac{\Gamma, A\left\{\frac{y}{x}\right\} \Rightarrow \Delta}{\Gamma, \exists x A \Rightarrow \Delta}$ y is fresh

$$(\exists \Rightarrow)_H \quad \frac{\left\{\Gamma, A\left\{\frac{t}{x}\right\} \Rightarrow \Delta \mid t \in cterms(\mathcal{L})\right\}}{\Gamma, \exists x A \Rightarrow \Delta}$$

$$(\Rightarrow \exists) \quad \frac{\Gamma \Rightarrow A\left\{\frac{t}{x}\right\}, \Delta}{\Gamma \Rightarrow \exists x A, \Delta} \; t \text{ is closed}$$

Sequent Calculus for Herbrand Logic $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}}$

Γ and Δ are finite sets of formulas

Proof trees are finite

$$(\forall \Rightarrow)\quad \frac{\Gamma, A\left\{\frac{t}{x}\right\} \Rightarrow \Delta}{\Gamma, \forall x A \Rightarrow \Delta}$$

~~$(\Rightarrow \forall)$~~ $\dfrac{\Gamma \Rightarrow A\left\{\frac{y}{x}\right\}, \Delta}{\Gamma \Rightarrow \forall x A, \Delta}$ y is fresh $\qquad (\Rightarrow \forall)_{IND}$ induction rule

~~$(\exists \Rightarrow)$~~ $\dfrac{\Gamma, A\left\{\frac{y}{x}\right\} \Rightarrow \Delta}{\Gamma, \exists x A \Rightarrow \Delta}$ y is fresh $\qquad (\exists \Rightarrow)_{IND}$ induction rule

$$(\Rightarrow \exists)\quad \frac{\Gamma \Rightarrow A\left\{\frac{t}{x}\right\}, \Delta}{\Gamma \Rightarrow \exists x A, \Delta}$$

Effective Systems: $\forall$

$$(\Rightarrow \forall)_H \ \frac{\left\{\Rightarrow A\left\{\frac{t}{x}\right\} \mid t \in cterms(\mathcal{L})\right\}}{\Rightarrow \forall x A}$$

$$\rightsquigarrow$$

$$(\Rightarrow \forall)_{IND} \ \frac{\left\{A\left\{\frac{x}{x}\right\} \Rightarrow A\left\{\frac{f(x)}{x}\right\} \mid f \in func(\mathcal{L})\right\}}{\Rightarrow \forall x A}$$

Example

$func(\mathcal{L}) = \{0, s\}$

$$\frac{\Rightarrow A\left\{\frac{0}{x}\right\} \quad A\left\{\frac{x}{x}\right\} \Rightarrow A\left\{\frac{s(x)}{x}\right\}}{\Rightarrow \forall x A}$$

Effective Systems: $\forall$

$$(\Rightarrow \forall)_H \ \frac{\left\{\Rightarrow A\left\{\frac{t}{x}\right\} \mid t \in cterms(\mathcal{L})\right\}}{\Rightarrow \forall x A}$$

$$\rightsquigarrow$$

$$(\Rightarrow \forall)_{IND} \ \frac{\left\{A\left\{\frac{x_1}{x}\right\}, \ldots, A\left\{\frac{x_n}{x}\right\} \Rightarrow A\left\{\frac{f(x_1,\ldots,x_n)}{x}\right\} \mid f \in func(\mathcal{L})\right\}}{\Rightarrow \forall x A}$$

Example

$func(\mathcal{L}) = \{0, s\}$

$$\frac{\Rightarrow A\left\{\frac{0}{x}\right\} \quad A\left\{\frac{x}{x}\right\} \Rightarrow A\left\{\frac{s(x)}{x}\right\}}{\Rightarrow \forall x A}$$

Effective Systems: equality (1)

$$(\Rightarrow=)\ \frac{}{\Gamma \Rightarrow t = t, \Delta}\, t \in \mathit{cterms}(\mathcal{L}) \quad \rightsquigarrow \quad (\Rightarrow=)_{IND}\ \frac{}{\Gamma \Rightarrow x = x, \Delta}$$

$$(\mathit{paramodulation})\ \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, s = t \Rightarrow, A', \Delta}\, s, t \in \mathit{cterms}(\mathcal{L})$$

Effective Systems: equality (2)

$$(\Longrightarrow)\ \frac{}{\Gamma, s = t \Rightarrow \Delta}\, s \neq t \in \mathit{cterms}(\mathcal{L})$$

$$\rightsquigarrow$$

$$(\Longrightarrow)_1\ \frac{}{\Gamma, f(x_1, \ldots, x_n) = g(y_1, \ldots, y_m) \Rightarrow \Delta}\, f \neq g \in \mathit{func}(\mathcal{L})$$

$$(\Longrightarrow)_2\ \frac{\Gamma, x_i = y_i \Rightarrow \Delta}{\Gamma, f(\ldots, x_i, \ldots) = f(\ldots, y_i, \ldots) \Rightarrow \Delta}$$

Effective Systems: equality (2)

$$(=\Rightarrow)\ \frac{}{s = t \Rightarrow} \; s \neq t \in \mathit{cterms}(\mathcal{L})$$

$$\rightsquigarrow$$

$$(=\Rightarrow)_1\ \frac{}{f(x_1, \ldots, x_n) = g(y_1, \ldots, y_m) \Rightarrow} \; f \neq g \in \mathit{func}(\mathcal{L})$$

$$(=\Rightarrow)_2\ \frac{x_i = y_i \Rightarrow}{f(\ldots, x_i, \ldots) = f(\ldots, y_i, \ldots) \Rightarrow}$$

Effective Systems: equality (2)

$$(=\Rightarrow)\ \frac{}{s = t \Rightarrow} s \neq t \in cterms(\mathcal{L})$$

$$\rightsquigarrow$$

$$(=\Rightarrow)_1\ \frac{}{f(x_1, \ldots, x_n) = g(y_1, \ldots, y_m) \Rightarrow} f \neq g \in func(\mathcal{L})$$

$$(=\Rightarrow)_2\ \frac{x_i = y_i \Rightarrow}{f(\ldots, x_i, \ldots) = f(\ldots, y_i, \ldots) \Rightarrow}$$

Example

$func(\mathcal{L}) = \{0, s\}$

$$(=\Rightarrow)_1\ \frac{}{0 = s(x) \Rightarrow}$$

Effective Systems: equality (2)

$$(=\Rightarrow)\ \frac{}{s = t \Rightarrow} s \neq t \in \mathit{cterms}(\mathcal{L})$$

$$\rightsquigarrow$$

$$(=\Rightarrow)_1\ \frac{}{f(x_1, \ldots, x_n) = g(y_1, \ldots, y_m) \Rightarrow} f \neq g \in \mathit{func}(\mathcal{L})$$

$$(=\Rightarrow)_2\ \frac{x_i = y_i \Rightarrow}{f(\ldots, x_i, \ldots) = f(\ldots, y_i, \ldots) \Rightarrow}$$

Example

$\mathit{func}(\mathcal{L}) = \{0, s\}$

$$(=\Rightarrow)_2\ \frac{x = y \Rightarrow}{s(x) = s(y) \Rightarrow}$$

Soundness Theorem

- $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}}$
- $\mathbf{G}^{\mathbf{IND}}_{\mathbf{sHer}_=} = \mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}} + (\Rightarrow=)_{IND} + (paramodulation)$
- $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}_=} = \mathbf{G}^{\mathbf{IND}}_{\mathbf{sHer}_=} + (=\Rightarrow)_1 + (=\Rightarrow)_2$

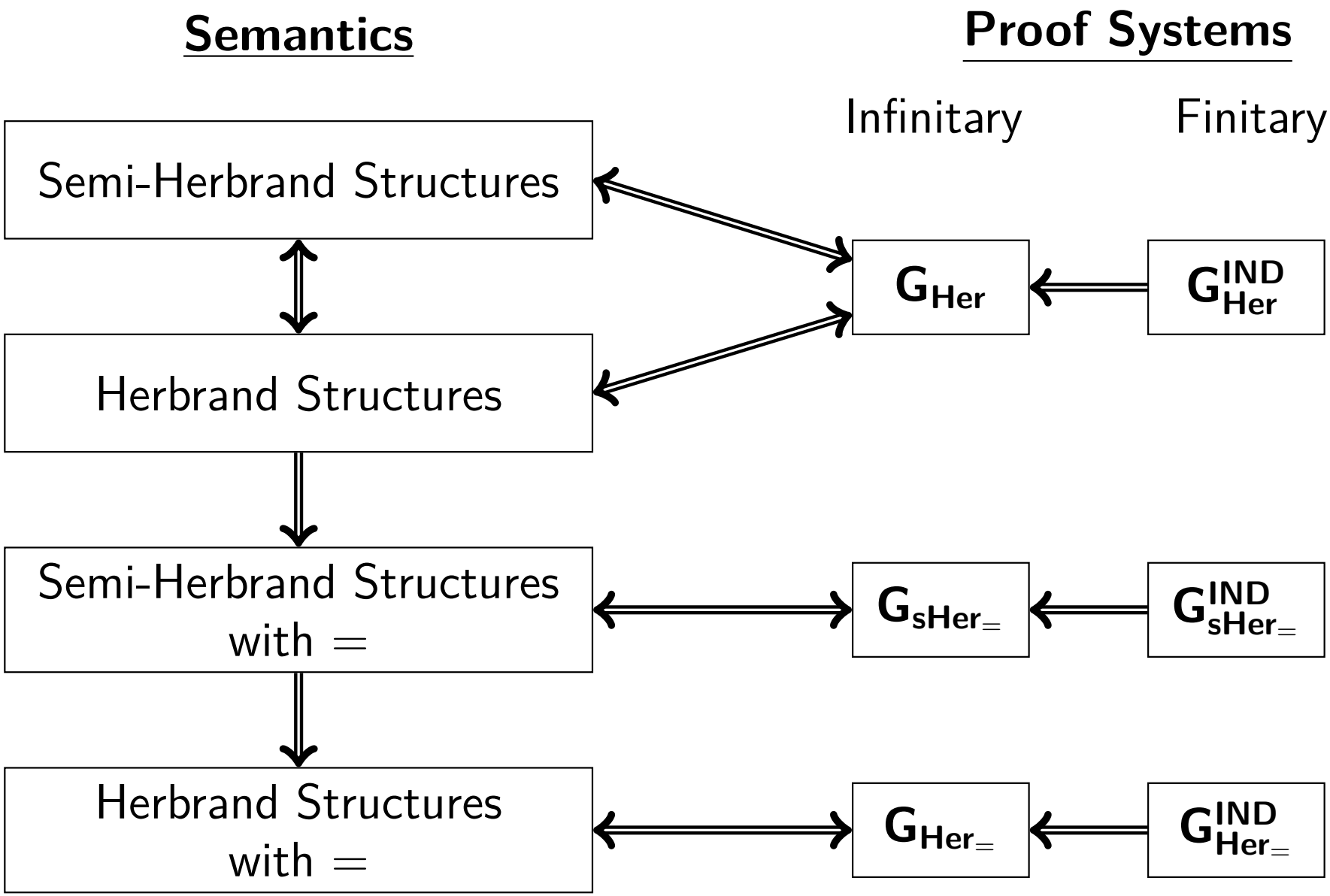
Theorem (soundness)

- $\mathcal{T} \vdash_{Her} A$ whenever the sequent $\mathcal{T} \Rightarrow A$ is derivable in $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}}$.
- $\mathcal{T} \vdash^{=}_{sHer} A$ whenever the sequent $\mathcal{T} \Rightarrow A$ is derivable in $\mathbf{G}^{\mathbf{IND}}_{\mathbf{sHer}_=}$.
- $\mathcal{T} \vdash^{=}_{Her} A$ whenever the sequent $\mathcal{T} \Rightarrow A$ is derivable in $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}_=}$.

Properties

- $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}}$, $\mathbf{G}^{\mathbf{IND}}_{\mathbf{sHer}_=}$ and $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}_=}$ are finite whenever $func(\mathcal{L})$ is
- Strictly stronger than **LK**
- Full $(=\Rightarrow)$ is derivable in $\mathbf{G}^{\mathbf{IND}}_{\mathbf{Her}_=}$ using $(=\Rightarrow)_1$ and $(=\Rightarrow)_2$

Herbrand Logics



Intuitionistic Herbrand Logic

- Herbrand logic carries definitional content
- Intuitionistic logic carries computational content
- Can the two be combined?

- An idea: Kripke models with Herbrand structures in each world
- Problem:
  - Domains are allowed to expand in intuitionistic logic
  - Domains are fixed in Herbrand logic
- Possible Resolutions:
  - Settle with CD (compromising intuitionism)
  - Semi-Herbrand structures (compromising Herbrandness)
  - A middle ground: Each world has a different language

Herbrand Logic and The Transitive Closure

Transitive Closure (TC)

- TC of a binary relation R is the minimal transitive relation that contains R
- TC Logic = FOL + TC Operator
- TC Logic is stronger than FOL and weaker than SOL

Example

Relation	TC
Sucessor	$<$
Edge	Path
Parent	Predecessor
Tomorrow	Future

Perhaps $TC \approx Herbrand$?

Herbrand Logic and The Transitive Closure

Herbrand Logic can express TC

Definition of helper relation qq:

$$\forall x.\forall z.(qq(x,z,0) \Leftrightarrow p(x,z))$$
$$\forall x.\forall z.(qq(x,z,s(n)) \Leftrightarrow qq(x,z,n) \vee \exists y.(qq(x,y,n) \wedge qq(y,z,n)))$$

Definition of q in terms of qq:

$$\forall x.\forall z.(q(x,z) \Leftrightarrow \exists n.qq(x,z,n))$$

But what about TC Operator?

TC can express "Herbrandness"

$$\forall w. \bigvee_{c \in const(\mathcal{L})} \left(\left(\bigvee_{f \in func(\mathcal{L})} y = f(x) \right)^* (c, w) \right)$$

But what about general functions?

Conclusions

We have seen:

- A modular definition of Herbrand semantics
- Infinitary proof systems
- Finitary approximations

Future work:

- Applications
- Proof theoretical properties
- Semantics for the effective systems
- Transitive Closure

Conclusions

We have seen:

- A modular definition of Herbrand semantics
- Infinitary proof systems
- Finitary approximations

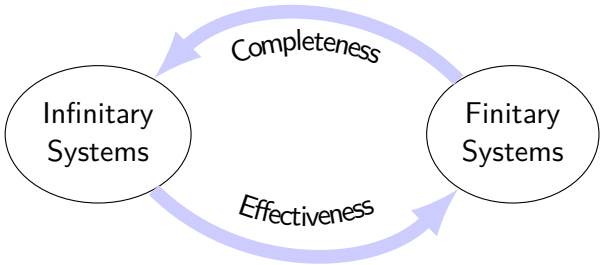
Future work:

- Applications
- Proof theoretical properties
- Semantics for the effective systems
- Transitive Closure

Thank you!

Proof Systems

- Infinitary proof systems
  - Consistency Proofs [Schutte'51,Pohlers'09]
  - Basis for practical subsystems [Feferman'62,Baker'93]
- Finitary approximations
  - Sound proof search
  - Algorithm for above applications



Automated Deduction in Arithmetic with the Omega Rule

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